

Supplier Performance Scorecard Normalization Framework

Current State Assessment, Market Benchmark, and Recommended Future-State Scoring
Model



Prepared for: Supplier Performance Management (SPM) Scorecard Redesign

Target: Q3 implementation planning

Version: Draft 1.0

Report Focus	Description
Current scorecard	Recreate and explain the existing penalty-based scorecard and its limitations.
Normalized scorecard	Define a scalable 0-100 supplier score based on applicable KPI weightage.
SVI layer	Position Supplier Value Index as a later benchmarking layer, not the immediate Q3 score.
Delivery approach	Phase the build from current-state trust building to future-state normalized scoring.

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1. Executive Summary

Recommended Q3 direction

Implement the BRD-aligned Normalized Supplier Score, 0-100, as the official Q3 scorecard. Use SVI, 0-1000, as a later leadership benchmarking layer after the normalized score is trusted and validated.

The current SPM scorecard is a penalty-based model: suppliers start at 100 and lose points based on deductions across KPI categories. This works for a fixed KPI structure, but it becomes unreliable when KPI applicability differs by supplier, zone, data source, or onboarding phase. The new framework should shift the scorecard from a fixed penalty model to an applicable-weight normalized model that remains explainable at every level.

Decision Area	Recommendation	Reason
Official Q3 score	Normalized Supplier Score, 0-100	Easy for business users to understand and directly aligned to BRD KPI pillars.
Formula approach	Earned KPI points divided by applicable KPI weightage	Allows fair comparison when suppliers have different KPI applicability.
Missing data handling	Show Coverage %, Coverage-Adjusted Score, and data-quality flags	Prevents incomplete data from looking like strong supplier performance.
SVI position	Phase 2 leadership benchmarking layer	Powerful for ranking, but should follow stakeholder validation of the normalized score.
Architecture	Metadata-driven configuration	Supports future KPI, zone, and threshold changes without code rewrites.

Executive takeaways

- The redesign is not just a UI refresh; it is a scorecard methodology and data-governance change.
- The current 100-minus-deductions model should be retained temporarily as a baseline and comparison view.
- The future score should normalize by applicable KPI weightage, not by a fixed assumption that every supplier has every KPI.
- Every score must be explainable through pillar score, KPI score, raw value, source system, refresh date, and coverage status.
- The Q3 release should prioritize trustworthy scoring and drill-down visibility before advanced percentile-based SVI ranking.

2. Current State Assessment

2.1 Current legacy score formula

Current SPM Score Formula

Current Supplier Score = 100 - Total Deductions

Total Deductions = Quality + Growth + Sustainability + DOT + IOT + POT + Risk

The current backend documentation describes this model as a deduction-based scorecard. Each KPI contributes penalty points, and the final score is capped so it cannot fall below zero.

2.2 Current KPI scoring pattern

KPI Area	Current Pattern	Observation
Quality	Raw quality inputs are converted into a Quality Score and then into a 0/10/20 deduction.	Useful but loses detail because many raw scores collapse into the same deduction bucket.
Sustainability	Scope 3 and Supplier Maturity convert into a Sustainability Score and then a 0/5/10 deduction.	Current logic is largely binary for Scope 3 data availability.
Operational	DOT, POT, and IOT contribute as independent deductions.	There is no consolidated Operational Pillar Score in the legacy model.
Risk	Risk score is bucketed into 0/10/20 deduction ranges.	Coarse treatment with only three final outcomes.
Growth	Innovation adopted divided by innovation shared, then mapped into deductions.	Manual or shared-drive dependency requires refresh governance.

2.3 What the current model does well

- It is simple to explain: the supplier starts at 100 and loses points for poor performance.
- It supports a single overall score and basic red/yellow/green interpretation.
- It is easier to validate when all suppliers have the same fixed KPI set.
- It provides a useful baseline for stakeholder walkthroughs before the new formula is introduced.

3. Why the Existing Scorecard Needs to Evolve

Current Limitation	Business Impact	Recommended Improvement
Fixed KPI assumption	Suppliers with different KPI availability are not fairly comparable.	Normalize by applicable KPI weightage.
Missing data can look like no penalty	Incomplete suppliers may appear better than suppliers with real but weaker data.	Separate Not Applicable from Missing Applicable and show coverage.
Bucketed deductions	A supplier at 91 and another at 100 can receive the same final deduction.	Retain raw values and score ratios; show granular KPI movement.
No pillar-level rollup in legacy model	Leadership cannot clearly see Service vs Operational vs Sustainability vs Value Creation contribution.	Calculate pillar scores before overall score.
Difficult future KPI onboarding	New zones and KPIs require bespoke logic changes.	Use metadata-driven KPI configuration.
Limited trend and root-cause visibility	Users see a score but not the reason behind score movement.	Add score-to-pillar-to-KPI-to-raw-data drill-down.

Core problem statement

The scorecard must fairly compare suppliers even when Supplier A has eight KPI scores, Supplier B has five, and Supplier C has only two applicable KPI scores. This requires normalization, coverage transparency, and configurable KPI logic.

4. External Market Benchmark

External supplier scorecard practices support the direction of weighted scoring, standardized KPI libraries, automated data capture, trend visibility, and transparent score decomposition. The goal is to move supplier performance management from static reporting to an action-oriented performance management process.

Market Reference	Relevant Practice	Implication for SPM
SAP Ariba SPM	Scorecards can include KPIs using survey and report data for defined evaluation periods.	The SPM model should support both system data and manual/stakeholder inputs.
Gartner supplier scorecard	Supplier performance improvement is linked to innovation, cost optimization, efficiency, and automated scorecards.	The scorecard should support supplier value creation, not only operational penalty tracking.
Ivalua supplier scorecards	Modern scorecards emphasize transparent KPI weighting, normalization, accountability, and signal-to-action workflows.	Normalization and actionability should be built into the design.
ISM supplier performance KPI guidance	Supplier performance measurement uses standardized KPIs, analytics, and improvement cycles.	The SPM score should drive governance and continuous improvement, not just a dashboard.

Market-backed design principle

A modern supplier scorecard should convert diverse supplier signals into a consistent, comparable, explainable score while preserving drill-down to the underlying raw values and actions.

5. Recommended Future-State Framework

5.1 Recommended Q3 scorecard model

Recommended Official Q3 Formula

Normalized Supplier Score =

$$\text{SUM}(\text{Pillar Score} \times \text{Pillar Weight}) / \text{SUM}(\text{Applicable Pillar Weight}) \times 100$$

Pillar Score =

$$\text{SUM}(\text{Earned Applicable KPI Points}) / \text{SUM}(\text{Applicable KPI Max Points})$$

This approach keeps the BRD business structure but improves the math underneath. Each KPI is scored using approved thresholds, rolled into a pillar score, and then scaled by the BRD pillar weight. If a KPI or pillar is genuinely not applicable, it is excluded from the denominator. If it is applicable but missing, it becomes a coverage issue rather than silently improving the supplier score.

5.2 Pillar structure

Pillar	BRD Weight	Role in Future Scorecard
Service Level	30	Measures delivery, quality, customer/supplier service, turnover, and PO acceptance performance.
Operational	25	Measures process performance such as price divergence, invoice conformity,

		and invoice timeliness.
Sustainability	25	Measures supplier maturity, Eclipse performance, and CO2/emission reduction progress.
Value Creation	20	Measures cost, cash, and engagement contribution.

5.3 Important weight normalization point

How to handle KPI max score mismatches

Some KPI max-score totals do not match the pillar weight directly. For example, Service Level KPI max scores total 45 while the pillar weight is 30. Therefore, KPI max scores should be used as relative weights inside the pillar, and the final pillar contribution should be scaled to the BRD pillar weight.

6. Calculation Methodology

Step	Calculation Action	Output
1	Capture raw KPI value from approved source system or manual file.	Raw KPI value with source and refresh timestamp.
2	Apply KPI direction metadata: higher-is-better, lower-is-better, or categorical mapping.	Comparable KPI orientation.
3	Apply BRD-approved scoring thresholds.	Earned KPI score and max KPI score.
4	Calculate KPI score ratio.	Earned score divided by max score.
5	Aggregate KPI scores into pillar score using applicable KPI weights.	Service, Operational, Sustainability, Value Creation scores.
6	Aggregate pillar scores into overall normalized supplier score.	Normalized Supplier Score, 0-100.
7	Calculate coverage and coverage-adjusted score.	Completeness and confidence layer.

6.1 Example normalization scenario

KPI	Max Score	Earned Score	Applicability
DOT	15	15	Applicable
Quality	5	3	Applicable
NPS	5	-	Not Applicable
Turnover	5	-	Not Applicable
PO Acceptance	15	10	Applicable

Example Pillar Score

$$\begin{aligned}
 \text{Service Pillar Score} &= (15 + 3 + 10) / (15 + 5 + 15) \\
 &= 28 / 35 \\
 &= 80\%
 \end{aligned}$$

6.2 Overall score example

Applicable-Pillar Normalization

If only Service and Operational are applicable:

$$\text{Overall Score} = ((\text{Service Score} \times 30) + (\text{Operational Score} \times 25)) / 55 \times 100$$

This prevents missing or non-applicable pillars from distorting the supplier score.

7. Coverage and Data Quality Layer

Normalization solves the fairness issue, but it can hide data incompleteness if used alone. Therefore, every supplier score should be accompanied by coverage and data-quality indicators.

Coverage Formulas

$$\text{Coverage \%} = \text{Available KPI Weightage} / \text{Total Expected KPI Weightage}$$

$$\text{Coverage-Adjusted Score} = \text{Normalized Supplier Score} \times \text{Coverage \%}$$

Data Scenario	Treatment in Scorecard
KPI is truly Not Applicable	Exclude from denominator; show N/A status.
KPI is applicable but data is missing	Flag as Missing; include in coverage gap; do not silently reward.
KPI value is invalid or out of range	Quarantine value, exclude from score for that period, and raise DQ alert.
Supplier has limited history	Calculate score but display Limited History badge.
New KPI added later	Add through KPI metadata and effective-dated configuration.
Manual KPI input	Require owner, refresh cadence, approval, and audit log.

Recommended header metrics

Display three values at supplier header level: Normalized Score, Coverage %, and Coverage-Adjusted Score. This makes the score useful while still showing how much confidence users should place in it.

8. Supplier Value Index (SVI) as a Later Maturity Layer

The SVI reference proposes a unit-less 0-1000 supplier index designed as a single comparable value across categories, zones, and spend sizes. It is positioned as the supplier equivalent of a customer lifetime value style index: one number that allows leadership to rank suppliers, while still enabling drill-down into pillar, KPI, and raw-value details.

SVI Formula

$$\text{SVI} = 1000 \times \text{SUM}(\text{Pillar Weight} \times \text{Pillar Percentile})$$

Pillar Percentile = Weighted average of KPI percentiles within the pillar

Every KPI is direction-normalized so higher = better before percentile ranking.

Dimension	Normalized Supplier Score	Supplier Value Index
Scale	0-100	0-1000
Best use	Official Q3 operational scorecard	Leadership ranking and benchmarking
Method	Threshold-based earned points and applicable weights	Percentile ranking across supplier base
Adoption risk	Lower because it aligns closely to BRD thresholds	Higher because percentiles require more stakeholder education
Recommendation	Implement first	Introduce after normalized score is validated

SVI positioning

Do not replace the Q3 normalized score with SVI immediately. Use SVI as a Phase 2 executive leaderboard once data quality, coverage, and business trust have matured.

9. UI and Deep-Dive Experience

9.1 Recommended scorecard layout

UI Area	Content to Display
Supplier header card	Supplier name, normalized score, coverage %, score band, last refresh date, data-quality status.
Pillar contribution view	Service, Operational, Sustainability, and Value Creation scores with weighted contribution.
KPI breakdown table	Raw value, earned score, max score, contribution, source, owner, refresh timestamp, status.
Trend view	Monthly, quarterly, and yearly movement by overall score, pillar, and KPI.
Raw data drill-down	Transaction-level or source-level data supporting each KPI score.
Comparison view	Current legacy score vs future normalized score, with explanation of differences.

9.2 Drill-down path

Recommended Drill-Down Hierarchy

```

Overall Supplier Score
  -> Pillar Score
    -> KPI Score
      -> Trend / Movement
        -> Raw Data / Transactions / Source Records
  
```

9.3 Stakeholder narrative flow

- Start by showing the current scorecard and reproduce the exact score to build trust.
- Explain where the current model works and where it becomes misleading.
- Introduce normalized scoring as a solution to supplier comparability and future KPI onboarding.
- Show coverage and data-quality flags so stakeholders understand score confidence.

- Preview SVI as a future leadership ranking layer rather than the first release model.

10. Data Model and Governance

10.1 Metadata-driven architecture

Entity	Recommended Fields
KPI Metadata	kpi_id, kpi_name, pillar, source_system, data_owner, direction, max_score, threshold_rules, scoring_method, applicability_rule, refresh_frequency, active_flag, effective_from, effective_to.
Supplier KPI Values	supplier_id, period, kpi_id, raw_value, unit, source_system, refresh_timestamp, applicable_flag, data_quality_status.
KPI Score Output	supplier_id, period, kpi_id, raw_value, earned_score, max_score, score_ratio, calculation_status, calculation_version.
Supplier Score Output	supplier_id, period, normalized_score, coverage_percent, coverage_adjusted_score, pillar_scores, weight_config_version.
Weight Configuration	config_version, pillar_weights, kpi_weights, approved_by, effective_date, change_reason.

10.2 Governance controls

- Maintain KPI thresholds and weights in configurable metadata rather than hardcoded backend logic.
- Version every threshold or weight change and store the version against each monthly supplier score.
- Require data owner and refresh cadence for every KPI source.
- Separate Not Applicable, Missing, Invalid, and Valid statuses in the score engine.
- Create an Admin view for data quality exceptions, active KPI coverage, and weight configuration changes.

11. Q3 Implementation Roadmap

Phase	Goal	Key Deliverables
Phase 1 - Current Scorecard Rebuild	Build trust by reproducing current logic.	Legacy score calculator, formula trace, comparison against workbook/application, current limitation summary.
Phase 2 - Normalized Score Engine	Implement official Q3 scoring model.	Metadata-driven KPI rules, pillar scores, normalized score, coverage %, coverage-adjusted score.
Phase 3 - UI and Deep Dive	Make the score explainable.	Supplier header card, pillar decomposition, KPI drill-down, trends, raw-data view.
Phase 4 - Validation and UAT	Secure stakeholder sign-off.	Sample supplier validation set, score reconciliation, business approval, production readiness.
Phase 5 - SVI Preview / Next Iteration	Introduce leadership benchmarking concept.	SVI prototype, percentile ranking, leaderboard design, adoption decision.

11.1 Suggested acceptance criteria

- 100% of in-scope KPI calculations are implemented and traceable to approved rules.

- 100% of score outputs show source, refresh timestamp, applicability status, and calculation version.
- Normalized supplier score reconciles correctly for a stakeholder-approved validation set.
- Coverage and missing-data flags are visible in the UI and report output.
- Business users can drill from overall score to pillar, KPI, trend, and underlying raw data.

12. Risks, Decisions, and Open Questions

12.1 Key risks and mitigations

Risk	Impact	Mitigation
Data owners are distributed across multiple teams.	Discovery and validation may be delayed.	Run weekly KPI owner working sessions and maintain an open-item tracker.
KPI definitions vary by zone or source.	Scores may become inconsistent.	Centralize definitions in KPI metadata and require business sign-off.
Missing data distorts normalized results.	Supplier rankings may be misleading.	Show Coverage %, Coverage-Adjusted Score, and Missing Applicable flags.
Stakeholders may resist formula change.	Adoption risk.	First reproduce the current scorecard, then explain limitations and transition to the new formula.
Manual KPIs are not ready for automation.	Incomplete score coverage.	Allow manual inputs with owner, cadence, approval, and audit trail.
Sustainability KPI labels conflict between BRD and SVI reference.	Ambiguity in sustainability pillar design.	Confirm authoritative KPI labels and weights before build.

12.2 Decisions to lock before engineering build

- Confirm which 8 KPIs are in scope for Q3 and which 6 are out of scope.
- Confirm whether N/A should be excluded from denominator and whether Missing Applicable should count toward coverage gaps only.
- Confirm KPI-level thresholds, especially where BRD and historical calculation logic differ.
- Confirm whether thresholds and weights must be editable by Super Admin in Q3 or deferred to a later release.
- Confirm historical backfill period and whether prior months should be recalculated under the new methodology.
- Confirm drill-down granularity by KPI source system.

13. Appendix - Formulas, KPI Weights, and Source References

13.1 Formula summary

Formula	Definition
Legacy Score	100 - sum of KPI deductions.
KPI Score Ratio	Earned KPI Score / Maximum KPI Score.
Pillar Score	Sum of earned applicable KPI points / sum of applicable KPI max points.
Normalized Supplier Score	Sum of Pillar Score x Pillar Weight / sum of applicable pillar weight x 100.
Coverage %	Available KPI weightage / total expected KPI weightage.
Coverage-Adjusted Score	Normalized Supplier Score x Coverage %.
SVI	1000 x sum of Pillar Weight x Pillar Percentile.

13.2 BRD-aligned pillar and KPI structure

Pillar	Pillar Weight	KPIs / Internal Max Scores
Service Level	30	DOT 15; NPS 5; Quality 5; Turnover 5; PO Acceptance 15.
Operational	25	Price Divergence 10; Invoice Conformity 5; IOT 10.
Sustainability	25	Supplier Maturity 10; Eclipse Score 5; CO2 Emission Reduction 5.
Value Creation	20	Cost 10; Cash 5; Engagement 5.

13.3 Source references

Ref	Source	How used
I1	SPM BRD (1).docx	Future-state scope, pillar structure, KPI definitions, deep-dive requirements, and BRD weightages.
I2	SPM_CALCULATION_LOGIC.md	Legacy final score formula, deduction behavior, color logic, and backend calculation flow.
I3	SPM KPI Data Source.pdf	Current KPI source mapping and legacy threshold logic.
I4	StrategicKPI.pdf	Quality, Sustainability, and Growth measure definitions and business context.
I5	OperationKPI.pdf	DOT, POT, and IOT operational definitions, formulas, and ownership.
I6	SPM_New_download.xlsx	Sample workbook and current data structure for supplier/KPI calculations.
I7	SPM Supplier Value Index (SVI).pdf	SVI concept, percentile normalization, coverage handling, weight governance, and roadmap.
E1	SAP Help Portal - SPM scorecards, surveys, and KPIs	Market reference for scorecards with KPIs, surveys, report data, and evaluation periods.
E2	Gartner - Supplier Scorecard	Market reference for automated scorecards and supplier value/performance improvement.
E3	Ivalua - Supplier Scorecards	Market reference for transparent weighting, normalization, and signal-to-action supplier performance.
E4	ISM - Supplier Performance Measurement KPIs	Market reference for standardized KPIs, analytics, and improvement cycles.

13.4 Recommended next working session agenda

- Walk through current scorecard formula and reproduce one sample supplier score.
- Review current limitations with examples: missing data, bucketed deductions, and KPI applicability differences.
- Validate the normalized formula and coverage rules with business stakeholders.
- Confirm in-scope KPIs, source owners, refresh cadence, and data granularity.
- Agree on Q3 MVP scope and defer SVI leaderboard to Phase 2 unless stakeholders explicitly approve it for Q3.